HeatWise — Presentation Diagrams

For: Investor / Team Meeting | March 2026

DIAGRAM 1 — The Problem We Solve

```
INDIAN CITY ROOFTOP TODAY          WITH HEATWISE

┌──────────────────────┐
│  BARE CONCRETE       │   ──►   🌿🌱🪴🌾🌵🌺
│  45°C surface        │         🌿  GREEN GARDEN 🌾
└──────────────────────┘         🌵  22-28°C surface
                                 🌿🌺🪴🌱🌾🌵
```

Problems:
- Urban Heat Island
- High AC energy bills
- Air pollution + CO_2
- Dead unusable space
- No expert guidance

Results:
- ✅ -3.8°C average cooling
- ✅ -22% energy consumption
- ✅ 1.2T CO_2 offset/year
- ✅ Usable green space
- ✅ AI-generated plan in minutes

DIAGRAM 2 — 5-Step User Journey

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5
📱 SCAN	📋 TELL US	🤖 AI PLAN	🌿 3D VIEW	🔧 INSTALL
Point phone at your rooftop AR measures area in m²	Sun/water/ wind/budget details (2 min form)	Instant plant recommendation + layout generated in <10 sec	See your rooftop in 3D before installing	Certified installer quotes in 24 hours
[Camera AR]	[Smart Form]	[ML Engine]	[Three.js]	[Partner Net]

DIAGRAM 3 — AI Recommendation Engine

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                    USER INPUT
              (space, sun, water, budget)
                         |
                         ▼
        ┌──────────────────────────────┐
        │     LAYER 1: ML MODEL        │
        │   Python · Trained on        │
        │   install outcomes data      │
        │   Confidence: HIGH           │
        └──────────────┬───────────────┘
              Works? ──┤ Fails?
              │        |
              │        ▼
              │  ┌────────────────────┐
              │  │  LAYER 2: CATALOG  │
              │  │  HYBRID ENGINE (TS) │
              │  │  51 species × 8 trait│
              │  │  scoring dimensions │
              │  │  Confidence: MEDIUM │
              │  └──────────┬─────────┘
              │    Works? ──┤ Fails?
              │             |
              │             ▼
              │  ┌────────────────────┐
              │  │  LAYER 3: RULES-ONLY │
              │  │  Pure TypeScript   │
              │  │  Always available  │
              │  │  Confidence: LOW   │
              │  └────────────────────┘
              │             |
              └─────────────┘
                         |
                         ▼
              SCORED PLANT LIST
```

- Heat Reduction Estimate
- Confidence Rating

DIAGRAM 4 — Species Intelligence (51 Plants)

51 CATALOG SPECIES

HERBS (10)	ORNAMENTALS (8)	SUCCULENTS (7)
Tulsi holy	Bougainvillea	Sedum ★
Basil sweet	Marigold	Prickly pear ★
Mint	Portulaca	Adenium ★
Coriander	Zinnia	Aloe vera
Curry leaf	Crossandra	Jade plant
Lemongrass	Duranta	Caladium
Brahmi	Vinca	—
Fenugreek	Chrysanthemum	
Geranium	—	
—		

VEGETABLES (8)	FOLIAGE (7)	GRASSES (4)
Cherry tomato	Pothos ★	Vetiver ★★
Chilli	Snake plant ★	Lemongrass ★★
Malabar spinach	Dracaena	Dense lemongrass
Okra	Coleus	Bamboo dwarf
Bitter gourd	Spider plant	
Eggplant	Areca palm	
Luffa	Wandering jew	
Malabar spinach		

★ = Drought-tolerant (scarce water scenarios)
★★ = High cooling score + wind resistant

DIAGRAM 5 — Platform Architecture

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┌──────────────────────────────────────────────────────────┐
│                        CLIENTS                           │
│   📱 React PWA     📱 Android App     💻 Admin Dashboard  │
└──────────────────────────┬───────────────────────────────┘
                           │  HTTPS
┌──────────────────────────▼───────────────────────────────┐
│                NEXT.JS API LAYER (60+ routes)            │
│  /recommendations  /reports  /installers  /geospatial    │
└──────┬──────────────────┬──────────────────┬─────────────┘
       │                  │                  │
┌──────▼──────┐   ┌───────▼───────┐  ┌──────▼──────┐
│ PRISMA ORM  │   │ RECOMMENDATION│  │ OPENAI API  │
│ SQLite/     │   │ ENGINE        │  │ GPT-4V      │
│ PostgreSQL  │   │ ML + Hybrid   │  │ Visuals     │
└─────────────┘   │ + Rules       │  └─────────────┘
                  └───────────────┘
```

DIAGRAM 6 — Business Model

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FREE USERS                    REVENUE STREAMS
__________                    _______________


User scans rooftop            1. INSTALLER COMMISSION
Gets AI recommendation    ──►    10-15% of install value
Views 3D garden plan           ┌─► Avg install: ₹80,000-2,00,000
                               │   Commission: ₹8,000-30,000/job
                               │
Request install quote     ──►  2. PREMIUM FEATURES
Certified installer           │   • Multi-project management
contacts in 24h               │   • Advanced AI visuals
                               │   • Irrigation planning
                               │
Post-install follow-up    ──►  3. DATA & INSIGHTS
                                   • Regional cooling data
                                   • Species performance
                                   • City planning partnerships
```

DIAGRAM 7 — Go-To-Market: Indian Cities

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                    TARGET CITIES (Phase 1)


TIER 1 METRO                      TIER 2 HOT CITIES
____________                      _________________

🏙 Delhi NCR        45°C+         🏙 Jaipur       46°C+
🏙 Mumbai           38°C+         🏙 Nagpur       46°C+
🏙 Bengaluru        34°C+         🏙 Lucknow      44°C+
🏙 Hyderabad        40°C+         🏙 Ahmedabad    45°C+
🏙 Chennai          38°C+         🏙 Indore       42°C+

Target property types:
┌──────────────┬──────────────────┬──────────────────┐
│  Residential │  Housing Society │   Commercial     │
│  Rooftops    │  (50-500 units)  │  Office/Retail   │
│  ₹60K-1.5L   │  ₹5L-50L project │  ₹10L-2Cr        │
└──────────────┴──────────────────┴──────────────────┘
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DIAGRAM 8 — Technical Moat

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WHAT COMPETITORS DO                WHAT HEATWISE DOES
____________________               ____________________


Generic green roof PDFs     →      Personalized AI plan in 60 sec
Manual site survey needed   →      AR measurement from your phone
One-size-fits-all plants    →      51-species catalog × 8 conditions
No data on outcomes         →      Immutable telemetry → ML improves
No installer network        →      Certified partner network + RFQ
Desktop-only tools          →      Mobile-first PWA + Android app
No post-install tracking    →      Longitudinal follow-up system
```

DIAGRAM 9 — Key Metrics (Current Demo)

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┌──────────────┐  ┌──────────────┐  ┌──────────────┐  ┌──────────────┐
│   51         │  │  -3.8°C      │  │  60 sec      │  │  3-layer     │
│  Species     │  │  Avg cooling │  │  Time to     │  │  AI fallback │
│  in catalog  │  │  prediction  │  │  recommendation│  │  (always on) │
└──────────────┘  └──────────────┘  └──────────────┘  └──────────────┘


┌──────────────┐  ┌──────────────┐  ┌──────────────┐  ┌──────────────┐
│  179         │  │  7           │  │  55+         │  │  100%        │
│  Species     │  │  Regression  │  │  API routes  │  │  Scenario    │
│  aliases     │  │  test cases  │  │  in platform │  │  pass rate   │
└──────────────┘  └──────────────┘  └──────────────┘  └──────────────┘
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Use these diagrams in presentations. Mermaid / draw.io versions available on request.